

VARAD PATIL

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in LLM systems, RAG architectures, and climate AI. Published researcher with experience building conversational AI, predictive forecasting systems, and scalable ML pipelines. Proficient in prompt engineering, Flask/FastAPI, and production AI deployment. Startup-ready mindset with ability to rapidly prototype and ship user-facing AI features. Pursuing M.Tech in AI at IIT Kanpur (B.Tech GPA: 9.70/10, GATE AIR 722).

EDUCATION

Indian Institute of Technology Kanpur — M.Tech in AI for Sustainability GPA: 9.25/10 2025–Present
Core Focus: Machine Learning, Deep Learning, Sustainable AI Architectures

Rajarambapu Institute of Technology, Maharashtra — B.Tech in Computer Engineering GPA: 9.70/10 2021–2025

RESEARCH PUBLICATION

A RAG-Based Legal Chatbot Using Llama2 and Advanced Information Retrieval
First Author — Proceedings of ICTCS 2024, Volume 6, Springer 2024

- Engineered production-ready legal chatbot with FAISS vector indexing and Llama2-7B via LangChain
- Designed corpus chunking, embedding strategies, and prompt templates for consistent response generation
- Built evaluation framework measuring precision, latency, and user satisfaction across 500+ test queries

PROFESSIONAL EXPERIENCE

Software Development Intern — Altair Engineering, Bangalore Jan–Jul 2025

- Automated HPC simulation workflows using Python pipelines, reducing manual effort by 40%
- Built REST APIs and MERN dashboards for real-time job monitoring and orchestration
- Deployed scalable data pipelines on AWS (S3, Athena, EC2) for large-scale simulation analytics
- Collaborated cross-functionally on system optimization initiatives, improving production reliability

Software Development Intern — Bubble Byte Ventures Pvt Ltd Sep 2022–Aug 2023

- Developed Python automation scripts improving system throughput by 30%
- Delivered production-ready web applications with rapid iteration cycles under startup constraints

AI/ML PROJECTS

Cyclone Track Prediction System *XGBoost, Gradient Boosting, Scikit-learn, Python*

- Built ML forecasting system predicting cyclone positions at 6h/12h/24h/48h/72h horizons; engineered 40+ features from IBTrACS dataset including motion vectors, intensity gradients, and spatial context
- Achieved 14.47km RMSE at 6h and 78.77km at 24h; designed evaluation pipeline with geodesic metrics and skill score analysis vs persistence baseline

Active-Break Indian Summer Monsoon Forecasting *PyTorch, TCN, Transformers, XGBoost*

- Built deep learning framework predicting Active, Break, and Normal monsoon phases up to 7 days ahead using ERA5 reanalysis data and engineered atmospheric features
- Developed TCN-Transformer Meta-Ensemble; outperformed persistence and climatological baselines; aligned with M.Tech research in sustainable AI

Air Quality Early Warning System *Gradient Boosting, Scikit-learn, Flask, Python*

- Built next-day AQI forecasting system for Indian cities using temporal feature engineering and Gradient Boosting regression
- Integrated CPCB health-advisory classification for public health alerts; exposed via deployable REST API with frontend demo

AI Resume Analyzer *Gemini API, Flask, MongoDB, NLP*

- Built intelligent resume evaluation using Gemini API for semantic analysis, ATS scoring, and personalized recommendations
- Designed keyword matching and role-alignment ranking logic; implemented analysis history via MongoDB for persistent storage

TECHNICAL SKILLS

AI/ML: RAG Systems, LangChain, FAISS, Llama2, HuggingFace, Prompt Engineering, Vector Embeddings
Deep Learning: PyTorch, TensorFlow, Transformers, TCN, scikit-learn, Transfer Learning
LLM APIs: OpenAI, Gemini, Claude **NLP:** Semantic Search, Information Retrieval, STT, TTS
Climate AI: Time Series Forecasting, ERA5 Reanalysis, IBTrACS, Atmospheric Feature Engineering
Backend: FastAPI, Flask, REST APIs, Node.js **Databases:** MongoDB, PostgreSQL, Vector DBs
Cloud: AWS (S3, EC2, Athena), GCP **Tools:** Git, Docker, Postman **Frontend:** React.js, Next.js

ACHIEVEMENTS & LEADERSHIP

Community Lead — Google Developer Group (GDG) RIT 2024–2025
Led 45+ students; conducted technical workshops and mentored 20+ developers in ML and web technologies.

Key Achievements: GATE 2025 — All India Rank 722 | First-author Springer publication (ICTCS 2024)

Certifications: Generative AI — DeepLearning.AI Programming Foundations — Duke University